

UCC612: Railway & Airport Engineering

L	T	P	Cr
3	1	0	3.5

Course Objectives: This subject aims to develop an understanding of the structural & geometrical design of various components of railway track & airport pavements as per the latest standards & software's.

Railway Engineering:

Permanent way specifications& design: Gauges in railway tracks, typical railway track cross-section, coning of wheels, Rails: Function of rails, requirement of rails, types of rail sections – comparison of rail types, length of rail, rail wear, rail failures, creep of rails, rail fixtures and fastenings, Sleepers: Functions and requirements of sleepers, classification of sleepers, comparison of different types of sleepers, spacing of sleepers and sleeper density, Ballast: Function and requirements of ballast, types, comparison of ballast materials, cross-section design of permanent way.

Geometric design: Alignment design, horizontal curves, super elevation, equilibrium, cant and cant deficiency, length & setting out of transition curve, gradients and grade compensation, use of OpenRail&HEADS Rail software.

Points and crossings: Introduction, necessity of points and crossings, design of a turnout as per Indian railways specifications.

Signaling and interlocking: Objects of signaling, engineering principle of signaling, classification of signaling, methods of interlocking, Traction and tractive resistances, stresses in track, Hauling capacity of locomotive and modernization of railway track.

Airport Engineering:

Runway orientation using wind rose diagrams, runway length & width, corrections for elevation, temperature & gradient, airport classification.

Runway & Taxiway Design: Geometric design of runway as per standard guidelines, taxiway layout, geometric design standards for taxiway & aprons, Rapid Exit Taxiways,

Structural design of runway pavements: Design of flexible and rigid runways as per FAA procedure using FAARFIELD & PCA method, Design of joints for airport pavements, Specifications for the different layers of runway and taxiway pavements, Pavement Evaluation for runway & taxiway, ACN-PCN method, ACR-PCR rating systems.

Pavement Evaluation & maintenance: COMFAA & ELMOD software's, Airport Pavement management systems for runway pavements.

Airport Layouts: Terminal area, parking area, apron & hanger typical airport layouts, Lightings and markings design for airside area of an airport.

Computational Software's:

1. FAARFIELD
2. COMFAA
3. ELMOD
4. ICAO-ACR
5. OpenRail

Assignment/Project:

1. To design the flexible & rigid runway & apron pavements.

2. To design the turnout as per the Indian Railway specifications.

Course Learning Outcomes(CLO):

Upon completion of this course, the students will be able to:

1. Determine the runway orientation and the runway length as per standard guidelines.
2. Design the airport pavements including air-side marking & lighting as per ICAO & FAA guidelines.
3. Evaluate pavements as per ICAO standards and learn the concept of pavement maintenance management system.
4. Employ Railway Track specifications and perform geometric design of the railway track.
5. Design of turnout and crossings as per the Indian Railways guidelines.

RecommendedBooks:

1. Rangawala, S. C., *Railway Engineering*, Charotar Publishers, Anand, (2002).
2. Arora, S. P. and Saxena., *Railway Engineering*, Dhanpat Rai Publishers, New Delhi, (2001).
3. Khanna, Arora and Jain. *Airport Planning and Design*, Nem Chand and Brothers, Roorkee, (2002).
4. *Railway Engineering*, Satish Chandra and M. M. Agrawal, Oxford University Press, New Delhi (2013).
5. Robert Horenjeff; *Planning and Design of Airports* (5th edition), McGraw Hill Book Co. (2010)

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1	MST	30
2	EST	40
3	Sessional (May include assignments/quizzes/tutorial evaluations)	30